

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
const int btnUP = 2;
```

```
const int btnDOWN = 3;
```

```
const int btnLEFT = 4;
```

```
const int btnRIGHT = 5;
```

```
byte head[] = {0x0E, 0x1F, 0x1F, 0x1F, 0x1F, 0x1F, 0x0E, 0x00};
```

```
byte body[] = {0x0E, 0x15, 0x1F, 0x1F, 0x1F, 0x15, 0x0E, 0x00};
```

```
byte food[] = {0x00, 0x0A, 0x04, 0x0E, 0x04, 0x0A, 0x00, 0x00};
```

```
int snakeX[20];
```

```
int snakeY[20];
```

```
int snakeLen;
```

```
int dirX;
```

```
int dirY;
```

```
int foodX;
```

```
int foodY;
```

```
int score;
```

```
unsigned long prevMillis = 0;
```

```
int gameDelay = 400;
```

```
bool running = false;
```

```
void setup() {  
    lcd.init();  
    lcd.backlight();  
    lcd.createChar(0, body);  
    lcd.createChar(1, head);  
    lcd.createChar(2, food);  
  
    pinMode(btnUP, INPUT_PULLUP);  
    pinMode(btnDOWN, INPUT_PULLUP);  
    pinMode(btnLEFT, INPUT_PULLUP);  
    pinMode(btnRIGHT, INPUT_PULLUP);  
  
    showStartScreen();  
}
```

```
void showStartScreen() {  
    lcd.clear();  
    lcd.setCursor(3, 0);  
    lcd.print("SNAKE PRO");  
    lcd.setCursor(0, 1);  
    lcd.print("Press Any Button");  

```

```
    while(digitalRead(btnUP) && digitalRead(btnDOWN) && digitalRead(btnLEFT) &&  
digitalRead(btnRIGHT));  
  
    initGame();
```

```
}
```

```
void initGame() {
```

```
    snakeLen = 3;
```

```
    score = 0;
```

```
    dirX = 1;
```

```
    dirY = 0;
```

```
    gameDelay = 400;
```

```
    for(int i = 0; i < snakeLen; i++) {
```

```
        snakeX[i] = 3 - i;
```

```
        snakeY[i] = 0;
```

```
    }
```

```
    spawnFood();
```

```
    running = true;
```

```
}
```

```
void spawnFood() {
```

```
    foodX = random(0, 16);
```

```
    foodY = random(0, 2);
```

```
}
```

```
void loop() {
```

```
    if (!running) return;
```

```
if (digitalRead(btnUP) == LOW && dirY == 0) { dirX = 0; dirY = -1; }  
if (digitalRead(btnDOWN) == LOW && dirY == 0) { dirX = 0; dirY = 1; }  
if (digitalRead(btnLEFT) == LOW && dirX == 0) { dirX = -1; dirY = 0; }  
if (digitalRead(btnRIGHT) == LOW && dirX == 0) { dirX = 1; dirY = 0; }
```

```
if (millis() - prevMillis >= gameDelay) {  
    prevMillis = millis();  
    moveSnake();  
}  
}
```

```
void moveSnake() {  
    int nextX = snakeX[0] + dirX;  
    int nextY = snakeY[0] + dirY;  
  
    if (nextX < 0) nextX = 15; else if (nextX > 15) nextX = 0;  
    if (nextY < 0) nextY = 1; else if (nextY > 1) nextY = 0;
```

```
for (int i = 0; i < snakeLen; i++) {  
    if (nextX == snakeX[i] && nextY == snakeY[i]) {  
        gameOver();  
        return;  
    }  
}
```

```
if (nextX == foodX && nextY == foodY) {
```

```
score++;  
snakeLen++;  
if (gameDelay > 150) gameDelay -= 20;  
spawnFood();  
}
```

```
for (int i = snakeLen - 1; i > 0; i--) {  
    snakeX[i] = snakeX[i - 1];  
    snakeY[i] = snakeY[i - 1];  
}
```

```
snakeX[0] = nextX;  
snakeY[0] = nextY;  
draw();  
}
```

```
void draw() {  
    lcd.clear();  
    lcd.setCursor(foodX, foodY);  
    lcd.write(2);  
}
```

```
for (int i = 0; i < snakeLen; i++) {  
    lcd.setCursor(snakeX[i], snakeY[i]);  
    lcd.write(i == 0 ? 1 : 0);  
}  
}
```

```
void gameOver() {  
    running = false;  
    lcd.clear();  
    lcd.setCursor(3, 0);  
    lcd.print("GAME OVER!");  
    lcd.setCursor(4, 1);  
    lcd.print("Score: ");  
    lcd.print(score);  
    delay(3000);  
    showStartScreen();  
}
```